

Human Structure Learning Outcomes

Introduction To Abdominal Cavity

Week 102

Gastrointestinal

Learning outcome

Define the anatomical position. ☐

Demonstrate the bony and cartilaginous landmarks visible or palpable on abdominal examination. ☐

Describe the organisation of the parietal and visceral peritoneum, and the principle organisation of the peritoneal cavity, including the lesser and greater sacs, mesenteries, and peritoneal 'ligaments'. ☐

Locate the stomach, liver, spleen, omenta, duodenum, jejunum and ileum of the small intestine, caecum, appendix, ascending, transverse, descending and sigmoid parts of the colon (and the rectum) within the abdomen using surface anatomy landmarks as well as descriptive abdominal regions and common incision sites. ☐

Identify the different parts of the small and large intestines, and describe their locations in the abdomen and list the anatomical differences between them. ☐

Identify abdominal organs on suitable radiographs or MRI scans and recognise common abnormalities. ☐

Session learning outcome:

Be able to identify the major intestinal organs and locate their surface landmarks.

Human Structure Learning Outcomes

Airways & Lungs

Week 103

Respiratory

Learning outcome

Describe the subdivisions of the thorax. ☐

List the structures of the thoracic wall including the thoracic spine, ribs, and sternum, and describe the articulations of these bones. ☐

Describe how the boundaries of the thoracic inlet and outlet are formed by the vertebrae, ribs, costal cartilages and sternum. ☐

Identify the thoracic vertebrae, ribs and sternum and the joint types between them. ☐

Label a diagram of the basic structure of the airway, summarising the anatomy of the bronchial tree and bronchopulmonary segments, and explain their functional significance in relation to inhalation injury. ☐

List the relations of the trachea and be able to identify them. ☐

Identify the features of each lung (and list the differences between left and right lungs). ☐

Draw and describe the topographical surface projections of the lungs and parietal pleural reflections of the thoracic wall. ☐

Identify the lobes and fissures of the lungs on suitable models or prosections and describe their major anatomical relationships to the ribs, heart, aorta and other major vessels of the thorax. ☐

Locate the impressions made in the left lung by the heart and the aorta. ☐

Identify the root of the lung and the structures that form it as they enter and leave the lung, and its hilum. ☐

Describe the lymphatic drainage of the lung and its medical significance (very briefly). ☐

Describe the nerve supply of lung structures (phrenic and vagus nerves, sympathetic fibres). ☐

Identify the trachea, bronchi, hila, lung fields and diaphragm on a chest radiograph. ☐

Session learning outcome:

To be able to identify the structures within the chest wall and link them to function and likely pathology.

Human Structure Learning Outcomes

Upper Gastrointestinal Tract

Week 104

Gastrointestinal

Learning outcome

Describe the location of the oesophagus as posterior to the trachea and anterior to the vertebral column, and recall its arterial supply, and venous and lymphatic drainage. ☐

List the stages of swallowing and be aware of the role of the muscles of the jaw, cheeks, lips and tongue within the oral cavity, the soft palate, pharynx, larynx and oesophagus during a swallow. ☐

Identify the key features of the stomach (position, parts, sphincter locations, blood supply, innervation, internal characteristics, and anatomical relationships with other abdominal organs in life). ☐

Describe the anatomy of the autonomic nervous supply of the abdomen and locate the sympathetic trunk, the coeliac ganglion and the coeliac and aortic plexuses. ☐

Explain referred abdominal pain, associating sites of pain with abdominal viscera. ☐

Describe the anatomy of abdominal lymphatic drainage and explain its significance in relation to spread of malignancy. ☐

Explain the role of imaging in capturing the patient swallowing. ☐

Session learning outcome:

Be able to identify the structure of the oesophagus and stomach and locate their surface landmarks.

Human Structure Learning Outcomes

Anatomy Of The Immune System

Week 105

Immunology and
microbiology

Learning outcome

Identify the components of the immune system and link them to function.



Session learning outcome:

To identify the components of the immune system and link them to function.

Human Structure Learning Outcomes

Thoracic Cage And Pleura

Week 106

Respiratory

Learning outcome

Demonstrate the surface anatomy landmarks associated with major thoracic structures. ☐

Describe the movements involved in normal, vigorous and forced ventilation and list the muscles responsible for these movements. ☐

Describe the functional anatomy of the intercostal muscles. ☐

Explain that the arterial supply of the thoracic wall is via intercostal vessels (as branches of the internal thoracic artery and aorta). ☐

Describe a neurovascular bundle in a typical intercostal space and outline the structures its components supply. ☐

Explain that the intercostal nerves are branches of the ventral primary rami of the thoracic spinal nerves and receive rami communicantes from the sympathetic trunk. ☐

Recognise normal structures on radiographs of the thorax in inspiration and expiration, and be able to recognise common abnormalities. ☐

Explain the concept of the pleural sac (parietal and visceral) and potential space. ☐

Identify the surface landmarks that are used when inserting a chest drain. ☐

Session learning outcome:

To identify the structures of the chest wall and link them to lung function.

Human Structure Learning Outcomes

Shoulder

Week 107

Musculoskeletal

Learning outcome

Identify and side a clavicle, identify its main anatomical landmarks on the bone and on suitable radiographs, and name its articulations and muscular and ligamentous attachments. ☐

Identify and side a scapula, identify its main anatomical landmarks, and name its articulations and muscular and ligamentous attachments. ☐

Explain that the sternoclavicular and acromioclavicular joints are atypical synovial joints, with fibrocartilage lining the articular surfaces. ☐

Identify the sternoclavicular (anterior and posterior), costoclavicular and coracoclavicular ligaments, and be aware that these strong ligaments mean that the clavicle rarely dislocates, and instead is the most commonly fractured bone. ☐

Locate the muscles: serratus anterior, pectoralis minor, pectoralis major, subclavius, latissimus dorsi, trapezius, levator scapulae, rhomboid minor, rhomboid major, deltoid, supraspinatus, infraspinatus, teres minor, teres major, and subscapularis. ☐

Describe the actions of the muscles acting on the shoulder girdle (scapula and clavicle) summarising their main attachments and somatic nervous innervation. ☐

Identify and side a humerus, identify its main anatomical landmarks and name its articulations and muscular and ligamentous attachments. ☐

Describe the glenohumeral (shoulder) joint as a large, synovial, ball-and-socket joint, identify the rotator cuff muscles, their actions, and recall the movements of this joint. ☐

Explain why the shoulder is the most commonly dislocated joint, recall that it dislocates inferiorly, and predict the functional and possible pathological consequences of this type of injury. ☐

Identify the boundaries and contents of the axilla. ☐

Label a diagram of the brachial plexus, and identify the parts of the plexus on a model or a prosection. ☐

Explain the loss of function resulting from injuries to the different parts of the brachial plexus. ☐

Recall the origins and functions of the axillary, radial, ulnar, musculocutaneous and median nerves in the arm, forearm, wrist and hand. ☐

Name the main muscles and muscle groups that the axillary, radial, ulnar, musculocutaneous and median nerves supply and their sensory distribution. ☐

Predict the consequences of common injuries to the axillary, radial, ulnar, musculocutaneous and median nerves and describe how to test their functional integrity. ☐

Session learning outcome:

To be able to identify the components of the shoulder joints and link them to function.

Human Structure Learning Outcomes

Heart

Week 108

Cardiovascular

Learning outcome

Identify the parts of the adult heart, both externally and internally, and explain their functional significance. ☐

Locate the atrio-ventricular, pulmonary and aortic valves on an intact heart, demonstrate their surface markings in the living and how they relate to clinical and radiological examinations and describe their role in the prevention of regurgitation of blood. You should also know where they can best be auscultated. ☐

Label the parts of the conducting system of the heart on a diagram. ☐

Describe the anatomical course of the spread of excitation through the chambers of the heart. ☐

Locate surface anatomy landmarks with respect to the heart (and great vessels), being aware of how they relate to clinical and radiological examinations including the placement of ECG electrodes. ☐

Describe the effects of sympathetic and parasympathetic innervation on the heart. ☐

Locate the cardiac plexus and recall that it contains both sympathetic and parasympathetic (from the vagus nerve) fibres. ☐

Explain the mechanism of 'referred pain' as it relates to myocardial ischaemia. ☐

Identify the right and left vagus nerves, right and left recurrent laryngeal nerves and sympathetic chains. ☐

Identify right, inferior and left heart borders and their constituent parts on a chest radiograph. ☐

Recognise chest radiographs that are taken in two ways: posteroanterior (PA) and anteroposterior (AP). ☐

Describe how the cardiothoracic ratio is determined on a PA chest radiograph. ☐

Session learning outcome:

To link the structure of the heart to its functions.

Human Structure Learning Outcomes

Introduction To Neuroanatomy

Week 109

Nervous

Learning outcome

Identify the rostral/anterior, caudal/posterior, inferior/ventral and superior/dorsal aspects of the brain and spinal cord. ☐

Identify the grey and white matter of the brain and spinal cord. ☐

Identify the cerebral hemispheres, corpus callosum, midbrain, pons, medulla, cerebellum, ventricles, thalamus, basal ganglia and hypothalamus on a 3D model/prosection of the brain and on MRI brain scans. ☐

Recognise the anatomy of a typical spinal nerve as exemplified by a thoracic spinal nerve, including its origin from the dorsal and ventral spinal roots. ☐

Recognise the parts of the developing nervous system and the processes involved in its development. ☐

Identify the frontal, parietal, temporal and occipital lobes. ☐

Identify and list the anatomical components of the central nervous system, peripheral nervous system and the autonomic nervous system. ☐

Identify the components of a spinal reflex arc and recognise the significance of enhanced and absent tendon reflexes. ☐

Identify the spinal root values of the five main reflexes (biceps, brachioradialis, triceps, knee and ankle). ☐

Session learning outcome:

To be able to identify the major parts of the nervous system and identify the processes involved in its embryological development.

Human Structure Learning Outcomes

Introduction To Embryology

Week 110

Reproductive

Learning outcome

Identify the structures that form during the development of the embryo and its supporting structures from fertilisation to 18 days, and recall the timings of the formation of these structures.

☐

Recall the timings of the embryonic and foetal periods, the relation of embryological timings to clinical gestation timings, and recognise the principles of growth and development.

☐

Session learning outcome:

To be able to identify the timings of the embryonic and foetal periods, the relation of embryological timings to clinical gestation timings, and recognise the principles of growth and development.

Human Structure Learning Outcomes

Ureter, Bladder And Urethra

Week 111

Renal

Learning outcome

Recognise common anatomical variants arising during the development of the urogenital system. ☐

Recall the relationships of the kidneys and ureters to other abdominal and pelvic structures and discuss the significance of these relationships with respect to urinary stones. ☐

Discuss micturition with regard to the consequences of obstruction of the ureter (kidney stones), and the urethra (common in elderly males due to enlargement of the prostate gland). ☐

Describe the course of the ureter with respect to other structures, and describe the pathway of urine from the renal pyramids to the urinary bladder. ☐

Recognise the bladder and its parts (including the ureteric openings), and identify the overlying peritoneum and the anatomical relationships of the bladder in male and female pelvises. ☐

Name the nerve fibres that innervate the bladder and its sphincters, recall their paths, and describe the mechanism of micturition. ☐

Recognise the anatomical parts of the urethra in males and females and link those parts to continence and catheterisation. ☐

Session learning outcome:

To identify the structures of the renal tract and link them to function.

Human Structure Learning Outcomes

Embryology Of The Gut

Week 112

Gastrointestinal

Learning outcome

Identify the parts of the developing gastrointestinal tract, and the processes by which this tract and associated structures develop and mature. ☐

Session learning outcome:

Identify the processes that form the embryonic gut tube, identify the parts of the developing gastrointestinal tract, and the processes by which this tract and associated structures develop and mature.

Human Structure Learning Outcomes

Hand, Wrist And Forearm

Week 113

Musculoskeletal

Learning outcome

Describe the anatomy of the superior and inferior radio-ulnar joints. ☐

Recognise the movements of supination and pronation and identify the muscles involved whilst summarising their main attachments and somatic nervous innervation. ☐

Describe and recognise movements at the wrist joint and identify the muscle groups responsible. ☐

Recognise the positions and functions of the retinacula of the wrist and the tendon sheaths of the wrist and hand, and relate them to carpal tunnel syndrome and the spread of infection in tendon sheaths. ☐

Describe the close relationships of the bones and joints of the forearm and wrist with other structures, e.g. blood vessels, nerves, ligaments, tendons and bursae, which may be injured in fracture and dislocation, and recognise the affects of such damage upon function. ☐

Describe the anatomical basis of the assessment of cutaneous sensation in the dermatomes of the upper limb and tendon jerk tests. ☐

Name and demonstrate the movements of the fingers and thumb. ☐

Identify the muscles and tendons involved in the movements of the fingers and thumb (differentiating between those in the forearm and those intrinsic to the hand) and name their attachments and nervous innervation. ☐

Recognise the main types of grip (power, precision and hook) and the roles of the muscles and nerves involved in executing them, and describe how muscle groups work in synergy to facilitate action. ☐

Label a diagram of the muscles and tendons of the palm of the hand. ☐

Label a diagram of the muscles and tendons of the dorsal hand. ☐

Locate the radial artery and superficial branch of the radial nerve. ☐

Locate the ulnar artery and ulnar nerve. ☐

Identify the bones of the wrist and hand and their relative positions, identifying those that are commonly damaged and be able to predict functional impairment following such damage. ☐

Identify the key features of the carpal, metacarpal and phalangeal bones. ☐

Draw a labelled diagram of the superficial and deep palmar arches describing the vascular supply of the hand and recognise major vessels in angiograms. ☐

Define the boundaries of the anatomical 'snuff box'. ☐

Human Structure Learning Outcomes

Session learning outcome:

To identify the structures of the hand, wrist and forearm and link them to function.

Human Structure Learning Outcomes

Female Reproductive System

Week 114

Reproductive

Learning outcome

Identify the mons pubis, parts of the female external genitalia, vestibule, urethral orifice and vaginal orifice. ☐

Identify the major viscera of the female pelvis. ☐

Describe the parts of the ovary, uterus and uterine tubes, and their anatomical relationships and function. ☐

Describe the arterial supply, and venous and lymphatic drainage of the ovary, uterus and uterine tubes. ☐

List or identify the structures attached to the uterus. ☐

Identify the levator ani and endopelvic fascia as the most important supports of the uterus (and recognise that both may be damaged during child birth). ☐

Identify the parts of the cervix, its anatomical relationships, and name the changes that occur within the uterus during pregnancy. ☐

Describe the arterial supply, and venous and lymphatic drainage of the vagina. ☐

Explain how the clitoris is analogous with the penis. ☐

Describe the anatomy of the superficial perineal pouch. ☐

Describe the innervation and mechanisms involved in erection of cavernous tissue. ☐

Session learning outcome:

To be able to identify the structures of the female reproductive system and link them to function and pathology.

Human Structure Learning Outcomes

Knee

Week 115

Musculoskeletal

Learning outcome

Identify and side a tibia, describing the main anatomical landmarks on bones and radiographs. ☐

Identify and side a fibula, describing the main anatomical landmarks and associations with the tibia on bones and radiographs. ☐

Identify a patella, locating attachments of quadriceps femoris and ligamentum patellae. ☐

Describe the movements of the knee joint. ☐

Label a diagram of the key ligaments and deep structures of the knee joint. ☐

Identify and describe the actions of the muscles at the knee joint (flexors and extensors) summarising their innervations and main attachments. ☐

Describe the factors that are responsible for the stability at the knee joint including the locking mechanism close to full extension of the lower limb. ☐

Explain what structures in close relation to the knee joint (including bursae) may be injured with trauma to the knee (including fractures, dislocation and ruptured cruciate ligaments). ☐

Locate the popliteal fossa and identify its boundaries and the structures within it. ☐

Discuss the functions and importance of joints. ☐

Recognise synovial, cartilaginous and fibrous joints, giving examples of each. ☐

Describe the basic structure, biomechanics, nerve and blood supply of synovial joints, and be able to label a diagram of such. ☐

Explain the significance of bursae and ligaments. ☐

Link the structure of the synovium with its function. ☐

Describe the processes involved in endochondral ossification and their roles in longitudinal bone growth and fracture healing. ☐

Session learning outcome:

To identify the structures of the knee joint and link them to function.

Human Structure Learning Outcomes

Introduction To The Brain

Week 119

Nervous

Learning outcome

Identify rostral, caudal, dorsal, ventral, left and right on images of an axial CT brain, an axial MRI brain and an axial MRI spinal cord. ☐

Identify the origins of the cranial nerves and identify their extra-cranial pathways to/from their effector/sensor structures and link to the functional outcomes of damage along those pathways. ☐

Identify the different structures in the brain stem (medulla, pons and midbrain), their relationships to each other and to the brain, spinal cord and other components of the posterior fossa, and link to the functional consequences of damage to these structures. ☐

Identify the main components of the cerebellum and its connection to the brain stem, spinal cord and cerebral hemispheres. ☐

Recognise the grey and white matter of the brain, the internal capsule, cerebral peduncles, and primary motor cortex. ☐

Session learning outcome:

To be able to identify the major parts of the brain and link them to function.

Human Structure Learning Outcomes

Hip

Week 125

Musculoskeletal

Learning outcome

Recognise the structure and describe movements of the hip joint (including the bones fused to create the acetabulum). ☐

Identify and explain the actions of the muscles on the hip joint, including flexors, extensors, abductors and medial rotators, adductors, lateral rotators. ☐

Describe the structures responsible for the stability of the hip joint and their relative contribution to maintaining the lower limb in different positions. ☐

Describe the sites of venous access that can be used for 'cut-down' procedures in emergencies. ☐

Describe the anatomy of the gluteal (buttock) region and the course of the sciatic nerve within it, explaining how to avoid it when giving intramuscular injections. ☐

Outline the origin of the lumbosacral plexus, the formation of its major branches and the structures they supply. ☐

Recognise the origin, course and function of the sciatic, femoral, obturator, common peroneal (fibular) and tibial nerves, sural and saphenous nerves and summarise the muscles and muscle groups that each supplies as well as their sensory distribution. ☐

Explain the anatomical basis for the loss of function resulting from common nerve injuries in the lower limb (sciatic, superior gluteal, tibial and common fibular nerves). ☐

Describe the dermatomes of the lower limb and outline the sensory nerve supply with respect to assessing spinal injuries. ☐

Outline the segmental values for movements and reflexes at the hip, knee, ankle and foot. ☐

Recognise structures of the hip and lower limb on radiographs. ☐

Session learning outcome:

To identify the major components of the hip joint and link them to function.

Human Structure Learning Outcomes

Pituitary And Thyroid

Week 126

Endocrine

Learning outcome

Locate the pituitary gland (hypophysis cerebri) within the sella turcica of the sphenoid bone. ☐

Label a diagram of the parts of the pituitary gland, namely the anterior lobe (adenohypophysis), pars anterior (pars distalis), pars intermedia, pars tuberalis and posterior lobe (neurohypophysis). ☐

Describe the anatomical relationships of the pituitary gland (diaphragma sellae, sphenoid, cavernous sinus, dorsum sellae, basilar artery, the pons). ☐

Identify the thyroid and parathyroid glands, noting the arterial supply and venous drainage of each and the relationship of the arteries with the laryngeal nerves and the strap muscles. ☐

Identify the processes by which the pituitary gland, thyroid gland and suprarenal glands develop and name the parts of these developing embryonic structures. ☐

Locate the pineal gland. ☐

Session learning outcome:

To identify the endocrine organs within the body, their structure, anatomical relation and embryological development.

Human Structure Learning Outcomes

Superficial Neck

Week 127

Nervous

Learning outcome

Demonstrate the locations of the sternum, clavicle, hyoid bone, mandible and mastoid process and hence define the boundaries of the neck. ☐

Identify key spaces of cervical fascia on a cross-sectional diagram, e.g. retropharyngeal, parapharyngeal, and carotid sheath. ☐

Describe the boundaries of the anterior triangle of the neck. ☐

Locate the carotid sheath and list its contents. ☐

Identify the surface landmarks that are used when inserting a central venous line. ☐

Describe the origins, courses and major branches of the common, internal and external carotid arteries. ☐

Locate the carotid pulse. ☐

Trace the paths of the vagus, accessory and hypoglossal nerves in the neck. ☐

Compare the internal and external jugular veins and list differences between them. ☐

Name the boundaries (and floor) of the posterior triangle of the neck. ☐

Describe the relationships of the scalenus anterior muscle. ☐

Locate the spinal accessory nerve within the posterior triangle, the positions of the roots and trunks of the brachial plexus, the external jugular vein and the subclavian vessels. ☐

Identify the sternocleidomastoid and trapezius muscles, and describe their actions and innervations. ☐

Identify the cervical plexus, and note its relationship to the accessory nerve. ☐

Identify the oesophagus and trachea noting their arterial supply and venous drainage. ☐

Label a diagram or a model of the arterial trunks that arise from the arch of the aorta in the thorax. ☐

Label a diagram or a model of the veins that drain into the brachiocephalic veins. ☐

Describe the lymphatic drainage of the neck (and head), identifying named lymph node collections and noting potential routes for the spread of malignant disease. ☐

Identify the phrenic nerves in the neck and recognise the consequences of surgical damage. ☐

Session learning outcome:

To identify the musculo-skeletal and vascular structures in the neck.

Human Structure Learning Outcomes

Respiratory Embryology

Week 128

Respiratory

Learning outcome

Name the parts of the developing respiratory tract, identify the processes by which this tract forms and matures, and identify the processes of the formation of all other structures of the respiratory system. ☐

Session learning outcome:

Name the parts of the developing respiratory tract, identify the processes by which this tract forms and matures, and identify the processes of the formation of all other structures of the respiratory system.

Human Structure Learning Outcomes

Cells And Infection

Week 130

Immunology and
microbiology

Learning outcome

Identify organelles and structures within the cell membranes and link them to function and likely pathology. ☐

Session learning outcome:

To be able to identify the structures within the cell and link them to both function and likely pathology.

Human Structure Learning Outcomes

Anatomy Of The Immune System

Week 131

Immunology and
microbiology

Learning outcome

Identify the components of the immune system and link them to function.



Session learning outcome:

To identify the components of the immune system and link them to function.

Human Structure Learning Outcomes

Pancreas And Gallbladder

Week 132

Gastrointestinal

Learning outcome

Describe the position, form and functional anatomy of the gall bladder and biliary tree; their relations in the abdomen and the significance of these relations to gall bladder inflammation and biliary stones. ☐

Describe the journey of bile from the liver to the gallbladder and from the gallbladder to the duodenum. ☐

Identify the head, neck, body and tail of the pancreas. ☐

Describe the relationships of the pancreas with other abdominal organs. Discuss the significance of these relationships with respect to pancreatitis and biliary stone disease. ☐

Recognise the main duct of the pancreas running the full length of the organ, beginning in the tail and ending in the duodenum at the major duodenal papilla with the bile duct (hepato-pancreatic ampulla of Vater). ☐

Recognise the arterial supply and venous and lymphatic drainage of the pancreas. ☐

Recognise the anatomy of the gall bladder and biliary tree on radiographic images and recognise biliary stones. ☐

Session learning outcome:

To be able to identify the structures within the pancreas and biliary tract and link them to function and likely pathology.

Human Structure Learning Outcomes

Spine

Week 133

Nervous

Learning outcome

Identify the main components of the spinal cord (ascending sensory tracts, descending motor tracts, dorsal and ventral roots, grey and white matter) and link to the functional consequences of damage at different points along the pathway. ☐

Describe the structures forming the vertebral column, its regions and functions. ☐

Label a diagram of a vertebra, and differentiate between atlas and axis, cervical, thoracic and lumbar vertebrae and sacrum. ☐

Explain the main anatomical features of vertebral bones on the bones themselves, on suitable radiographs and other images. ☐

Recognise the anatomy of intervertebral joints and intervertebral discs, and explain the role of the discs in weight bearing in the vertebral column. ☐

Describe common disc lesions that may impinge upon spinal nerves and/or the spinal cord. ☐

Explain the functions of the atlanto-occipital and atlanto-axial joints in the movements of the head, naming the main muscle groups involved. ☐

Identify the range of movement of the entire vertebral column and its individual regions. ☐

Identify the principal muscle groups and ligaments of the vertebral column and surface features in order to be able to perform an examination of the back. ☐

Discuss the functional role in stability and movement of the vertebral column and describe the anatomical basis of back pain. ☐

Recognise stable and unstable spinal injuries. ☐

List the anatomical relationships of the spinal cord and spinal nerves, with particular reference to the bones and joints of the vertebral column. ☐

Session learning outcome:

To be able to identify the components of the spinal cord and nerve roots and their anatomical relationships.

Human Structure Learning Outcomes

Coronary Arteries

Week 134

Cardiovascular

Learning outcome

Describe the course of the ascending aorta, the arch of the aorta and the descending thoracic aorta, naming the major branches of each division. ☐

Describe the origins, courses and main branches of the left and right coronary arteries and discuss the functional consequences of their obstruction. ☐

Explain the venous drainage of the heart. ☐

Recognise a coronary angiogram. ☐

Define the term 'anastomosis'. ☐

Describe the functions of the pericardium. ☐

Describe the layout of the pericardium (in relation to cardiac tamponade) and its sinuses. ☐

Explain the concept of pericardial sac and potential space. ☐

Recall that the phrenic nerves supply sensation from the pericardium. ☐

Explain that pain from the pericardium (pericarditis) may be referred to the shoulder tip because of the common innervation (C3, 4, 5). ☐

Recognise soft tissue structures on radiographic images of the thorax. ☐

Recognise coronary arteries on CT reconstruction images. ☐

Recognise the pericardium and associated structures and spaces on radiographs. ☐

Session learning outcome:

To be able to identify coronary arteries and link them to function and likely pathology.

Human Structure Learning Outcomes

Kidneys And Suprarenal Glands

Week 135

Renal

Learning outcome

Identify the external features of the kidney and explain its function. ☐

Recognise the gross internal features of the kidneys. ☐

Draw and label diagrams of coronal and transverse sections of the kidney through the hilum. ☐

List the anterior visceral relationships of both kidneys. ☐

List the autonomic nervous and arterial supply, and venous and lymphatic drainage of the kidney. ☐

Describe the adrenal glands' form, relations and functions. ☐

List the autonomic nervous and arterial supply, and venous and lymphatic drainage of the adrenal glands. ☐

Recognise the kidney in normal radiographs and recognise other methods of visualising the kidneys and urinary system. ☐

Identify the muscles of the posterior abdominal wall and explain their actions (psoas major, psoas minor, quadratus lumborum). ☐

Describe the origin, course and major branches of the abdominal aorta. ☐

Construct a labelled diagram of the caval venous system, and identify the important anastomoses. ☐

Identify the surface markings of the abdominal aorta and the caval venous system. ☐

Describe the location of the kidneys within the abdomen. ☐

Session learning outcome:

To be able to identify the structures within the kidney and link them to function and likely pathology.

Human Structure Learning Outcomes

Skull And Teeth

Week 136

Reproductive

Learning outcome

Label a diagrammatic sketch of the structure of a tooth. ☐

Recall the order and timings of tooth eruption. ☐

Describe the arrangement of the dentition. ☐

Identify the parts of the mandible. ☐

Name the boundaries of the temporal fossa. ☐

Identify and describe the temporalis muscle and locate the auriculotemporal nerve and the superficial temporal blood vessels. ☐

Identify the temporomandibular joint, the components of the joint and the accessory ligaments (temporomandibular, sphenomandibular, stylomandibular). ☐

Recall the movements that can occur during chewing and name the muscles involved in each movement. ☐

Explain what occurs in anterior temporomandibular joint dislocation and relocation. ☐

Name the muscles of mastication, their origins and insertions, actions and innervation (masseter, temporalis, lateral and medial pterygoid muscles). ☐

Locate the infratemporal fossa and name its boundaries. ☐

Recognise the major bones of the skull, including the frontal, parietal, occipital, temporal, maxilla, zygoma, mandible, sphenoid, nasal and ethmoid bones and identify their surface and imaging landmarks. ☐

Identify the major sutural joints of the adult skull and the fontanelles of the foetal skull. ☐

Identify the external and internal features of the cranial foramina of the skull, the bones through which they pass, listing the structures that each transmits. ☐

Describe the blood supply and nervous innervation of the teeth. ☐

Session learning outcome:

To be able to identify the structures of the face and teeth and link them to function.

Human Structure Learning Outcomes

Pharynx And Larynx

Week 139

Cardiovascular

Learning outcome

Locate the hard and soft palates, describing their muscular actions and bony components. ☐

Describe the actions of the muscles that move the soft palate and their nerve supply. ☐

Describe the innervation of the hard and soft palates (sensory and motor). ☐

Describe the components of the gag reflex. ☐

Describe the general disposition and relationships of the pharynx. ☐

Identify the opening of the auditory (Eustachian) tube in the nasopharynx ☐

List the constrictor muscles of the pharynx. ☐

Describe the mucosa, fascia and tissue spaces of the pharynx and explain the condition quinsy. ☐

Explain the innervation of the pharynx. ☐

List the structures passing through the walls of pharynx. ☐

Describe the cricopharyngeal sphincter (or upper oesophageal sphincter). ☐

Recognise the anatomical arrangement of the lymphoid tissue of the pharynx (Waldeyer's tonsillar ring). ☐

Identify the hyoid bone, the cartilaginous skeleton and ligamentous structures of the larynx. ☐

Demonstrate the palpable positions of the hyoid bone, thyroid and cricoid cartilages. ☐

Describe the muscles that elevate the larynx and facilitate closure of the inlet. ☐

Describe the intrinsic laryngeal muscles responsible for controlling vocal cord position and tension and how these function during phonation, laryngeal closure, the cough reflex and regulation of intrathoracic pressure. ☐

Recognise the chambers of the larynx. ☐

List the functions of the larynx. ☐

Describe the origin, course and functions of the motor and sensory nerve supplies of the larynx and the functional consequences of injury to these nerves. ☐

Recall and recognise the arterial supply, venous and lymphatic drainage of the larynx. ☐

Demonstrate on a model how the larynx works during breathing and phonation, noting that it opens during breathing and is partially closed during speaking. ☐

Recognise structures of the larynx and pharynx on radiographs. ☐

Human Structure Learning Outcomes

Session learning outcome:

To be able to identify the larynx and pharynx and link their structure to function and pathology.

Human Structure Learning Outcomes

Male Reproductive System

Week 141

Reproductive

Learning outcome

- Describe the anatomical features of the male external genitalia. ☐
- Identify the components of the male internal genital organs, recall their functions and anatomical relationships, and predict sites of referred pain from pelvic viscera. ☐
- Label a cross-sectional diagram of the penis taken at any level. ☐
- Identify the dorsal surface of the penis. ☐
- Describe the innervation, arterial supply, venous and lymphatic drainage of the penis. ☐
- Explain the anatomy of the perineal membrane and superficial perineal pouch in relation to the accumulation of fluids in the male. ☐
- Identify the scrotum and describe its structure, attachments, innervation, arterial supply, venous and lymphatic drainage. ☐
- Identify structures within the spermatic cord. ☐
- Explain the structure of the tunica vaginalis testis. ☐
- Describe the structure of the testis, epididymis and their coverings. ☐
- Describe the autonomic nervous, arterial supply, venous and lymph drainage of the testis and epididymis. ☐
- Explain the significance of the arterial supply of the testis in relation to torsion, the venous drainage in relation to varicocoele and the lymphatic drainage in relation to tumour spread. ☐
- Describe the structure and function of the prostate. ☐
- Describe the normal form of the prostate when examined per rectum and changes in relation to hypertrophy and malignancy. ☐
- Describe the innervation and mechanisms involved in erection of cavernous tissue and emission and ejaculation. ☐
- Trace the path of sperm and fluids through the male reproductive tracts. ☐
- Describe the path of the male urethra and recognise its three parts (prostatic, membranous and spongy), and the position of the external urethral sphincter. ☐
- Explain the functions of the bulbourethral glands. ☐

Session learning outcome:

To be able to identify the structures of the male reproductive system and link them to function and pathology.

Human Structure Learning Outcomes

Orbit

Week 142

Nervous

Learning outcome

Recognise the bones that comprise the orbit. ☐

Locate the superior orbital fissure and name the structures that pass through it. ☐

Name the muscles of the orbit, their actions and innervations. ☐

Identify the branches of the ophthalmic division of the trigeminal nerve and recognise their functions. ☐

Describe the anatomy of the optic nerve and chiasm. ☐

Describe the arterial supply and venous drainage of the orbit. ☐

Describe the origins, courses and functions of the oculomotor, trochlear and abducent nerves and predict the consequences of injury to them. ☐

Describe the anatomical relations of the cavernous sinus and common pathologies relating (anatomically) to it. ☐

Identify the ciliary ganglion and its branches and recognise consequences of pathology to it and its branches. ☐

Recall the route of the visual pathways from the eye to the occipital lobe and link to the functional consequences of damage at different points along that pathway. ☐

Identify the parts of the eye. ☐

Session learning outcome:

Identify the structures of the orbit and link them to function.

Human Structure Learning Outcomes

Small Intestine

Week 145

Gastrointestinal

Learning outcome

Describe the duodenum, its parts, position, secondary retroperitoneal attachment, blood supply and key relations with other abdominal organs and their significance in relation to peptic ulcer disease. ☐

Identify and differentiate between the jejunum and ileum (internally and externally). ☐

Label a diagram of the blood supply of the small intestine. ☐

Summarise the functional anatomy of the small intestine's mesentery; its structure, location and vascular, lymphatic and neural content. ☐

Recall that the lymph of the small intestine drains to the coeliac (duodenum) and superior mesenteric (duodenum, jejunum and ileum) lymph nodes. ☐

Describe the anatomy, innervation and functions of the muscles of the anterior and posterior abdominal walls. ☐

Discuss the functional relationships of the muscles of the anterior and posterior abdominal walls with the diaphragm and roles in posture, ventilation and voiding of abdominal/thoracic contents. ☐

In relation to direct and indirect inguinal hernias, demonstrate the anatomy of the attachments of the inguinal ligament, the anatomy of the superficial and deep inguinal rings and how the anterior abdominal wall muscles form the inguinal canal. ☐

List the contents of the inguinal canal in both males and females. ☐

Describe the relationship between the femoral canal and the inguinal ligament and the anatomy of femoral hernias. ☐

Session learning outcome:

To be able to identify parts of the small intestine and the structures related to the mesentery.

Human Structure Learning Outcomes

Mediastinum

Week 146

Respiratory

Learning outcome

Identify the parts of the developing somite, the processes by which the somites form, and the subsequent processes by which the somites contribute to the formation of the musculoskeletal structures of the embryo. ☐

Describe the courses and major relationships of the jugular and subclavian venous systems, and subclavian and carotid arteries. ☐

Describe the arrangement and contents of the superior, anterior, middle and posterior mediastina. ☐

Describe the differences between the left and right sides of the mediastinum. ☐

List the anatomical features that occur on the plane of the sternal angle of Louis and recognise why this is an important surface landmark. ☐

Describe the origins, courses and relationships of the brachiocephalic veins and superior vena cava and recognise the azygos system of veins. ☐

Describe the course and major relationships of the oesophagus within the thorax. ☐

Describe the origins and courses of the vagus and phrenic nerves in the thorax on both sides. ☐

Describe the autonomic innervation to the thoracic viscera, and describe the anatomy of the sympathetic chains and splanchnic nerves, including their composition and function. ☐

Explain the mechanism of referred pain from thoracic organs. ☐

List key collections of lymph nodes that drain the thorax. ☐

Describe the course and major relationships of the thoracic duct and the other lymphatic systems within the thorax, explaining their medical significance. ☐

Identify major thoracic structures on standard diagnostic images and be able to recognise common abnormalities. ☐

Session learning outcome:

To be able to identify normal structures within the mediastinum and link these to pathological abnormalities encountered in patients with lung cancer.

Human Structure Learning Outcomes

Embryology Of The Cardiovascular System

Week 147

Cardiovascular

Learning outcome

Name the parts of the developing embryonic heart and major blood vessels, identify the processes by which these structures form, and the differences between foetal and neonatal circulatory systems, and be able to recognise the symptoms of maladaptation of the circulatory system at birth. ☐

Session learning outcome:

To name the parts of the developing embryonic heart and major blood vessels, identify the processes by which these structures form, and the differences between foetal and neonatal circulatory systems.